

Haptic Sports Garment which can sense the player's every move and also analyse the data to inform the wearer of areas that need improvement. By measuring vibrations in key muscle groups, the garment will provide guidance to improve posture and even help maintain optimal speeds.

Wearable electronics is found to be especially useful for sportsmen recuperating from injuries, as it helps to track their improvement and even guides their movement using techniques like muscle emulation, video analysis, real-time suggestions and so on.

Spectator comfort and safety

Technology also plays a huge role in stadium design for good playing conditions as well as for assuring spectator comfort and safety.

Light-emitting diode (LED) based lighting systems with automatic control and monitoring, energy management systems, perimeter security tools and so on are widely used nowadays. The nature-inspired design of the Beijing National Stadium is one beautiful example.

Based on the design of a bird's nest, the stadium has several metres of unwrapped steel mesh that allows natural ventilation of the stadium. The temperature inside the stadium is regulated with the help of underground pipes that act as a geothermal heat pump system through which ground-source energy is collected. In winter the system absorbs the heat from the soil and helps heat the stadium, while in summer the coldness of the soil is used to reduce the temperature. The stadium is known for its sustainable design.

Last year, the New Meadowlands Stadium in New Jersey showcased a host of technologies designed in association with Cisco Connected Sports and Verizon to customise the look and feel of the venue according to the game. This is achieved using more than 2200 high-definition (HD) video displays and scoreboards located across the venue, and delivering custom digital video and

Staying in touch with fans

Team managers, coaches, players, match organisers, everybody is now on the social Web. They are making excellent use of social-networking tools to stay in touch with the public. Although Twitter, Facebook or USTREAM are obvious options, other applications like Venuing and Twackle are also being used to enhance the rapport between teams and fans.

According to some industry watchers, the social networking fever has another side to it. It is also helping players to earn some small change. Chad OchoCinco Experience, for example, is an iPhone application that provides a behind-the-scenes look into the life of the popular footballer Chad OchoCinco. Such applications enable athletes to earn money from both point-of-sale and ad revenue. Similarly, websites like fanwave.tv—a social network for fans—apparently pay athletes and coaches according to their engagement, which is measured by the number of 'tweets' they post.

wireless content to sports fans.

In India too, ticketing, crowd control and security solutions are being widely deployed in stadia. "There are quite a few technology solutions available for crowd control, surveillance, ticket counterfeiting detection, unauthorised entry detection, etc. Today, stadium security is important as there will be thousands of spectators and players whose security and convenience is of paramount importance," adds Prabhu.

Technology is also used for preparing the ground, checking the weather conditions and so on.

Enjoy the game, even at home

A discussion on technology in sports will be incomplete without a mention of the leading-edge broadcasting solutions being used by various television channels.

Apart from high-definition and three-dimensional broadcasts that are being pioneered by some channels, many direct-to-home service providers are also offering the facility of viewing the match from various angles. We have already discussed tools like Hawk-Eye and Hot Spot which are used to increase the efficacy of replays.

Nowadays, television is not the only medium to watch matches. With high-speed Internet connections and various Web video channels, you can watch the matches even on your computer. Companies such as SportsMechanics have advanced solutions that help the match organisers to relay their

matches effectively on the Internet. This enables even small matches to be relayed live.

Need to strike a balance

It is evident that a real lot of technology is available for sports. In fact, it is everywhere on the training ground. Yet, in a match, only certain technology is allowed, depending on the discretion of the organisers. All the discussed applications seem to have a positive effect on the game, so why should organisers be wary of allowing too much technology use by players?

One very recent example is the effect of the recent Union Cycliste Internationale (UCI) ban on radios. Radios on cycles have been used by athletes since 1990 for communicating with the coach. However, the UCI now feels that the use of radios has almost destroyed the cyclists' ability to think by themselves, and is increasing the reliance on the coach's instructions. That does not sound like a fair game, does it? Hence UCI decided to ban radios, but is faced with severe opposition from the cyclists who are refusing to drop their radio kits.

The fact is that too much of technology can kill a game's spirit and make it too mechanical; and when the organisers realise that the spirit is gone, it might be too late to ban the technology. ●

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